ALEJANDRO LUIS ARMAS BRAITHWAITE

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EDUCATION

University of Chicago — Chicago, IL
M.S. in Computational Analysis and Public Policy — GPA: 3.59/4.0 — **August 2026**
Relevant Coursework: *Program Evaluation (Causal Inference), Advanced Statistics I–II, Machine Learning, GIS*

Pacífico Business School — Lima, Peru
M.A. in Economics — **December 2019**
Coursework: *Advanced Econometrics, Applied Econometrics, Advanced Micro- & Macroeconomics*

Universidad del Pacífico — Lima, Peru
B.A. in Economics — Honors: Top Third of Students in 2015 and 2016 — **March 2018**

RESEARCH EXPERIENCE

The University of Chicago – Policy Lab (with IOM) — Chicago, IL
Policy Lab Consultant | Client: IOM — **August 2025 – December 2025**

- Co-developed (5-person team) a standardized, Sendai-aligned Excel cost model estimating ~US$395M in annual investment needs for gender- and migration-inclusive Disaster Risk Management (DRM) across 5 South American countries.
- Adjusted GDP benchmarks using 4 normalized IDB indices, allocated costs across 5 policy dimensions, and validated with IOM.

Detecting Political Ideology in News Media Articles — Chicago, IL
Research Project, in collaboration with a faculty researcher — **Summer 2025**

- Collected ~60,000 news articles and images (~10K per outlet) from 6 news media outlets using APIs, Selenium, and Requests.
- Cleaned and structured the multimodal (text + image) dataset for downstream NLP and computer-vision classification of political bias.

RESEARCH PROJECTS

Chicago Livability Score Project — Chicago, IL
University of Chicago — **Winter 2025**

- Co-designed a weighted livability index for 60 Chicago ZIP codes, building crime, school, and environment sub-indices from 257K records.
- Mapped point-level records to ZIP codes via spatial joins (GeoPandas, Shapely) and wrote pytest unit tests for a 4-person team's web map.

Predicting Dangerous Streets with ML — Chicago, IL
University of Chicago — **Spring 2025**

- Compared logistic regression, decision tree, random forest, and MLP models on NYC crash data, with SMOTE for class imbalance.
- Tuned Random Forest reached a 0.59 test F1 (vs. 0.36 random baseline, 0.49 logistic regression) by capturing non-linear feature interactions.

TEACHING EXPERIENCE

The University of Chicago – Harris School of Public Policy — Chicago, IL
Teaching Assistant – Machine Learning for Public Policy — **August 2025 – December 2025**

- Led TA sessions, held office hours, and graded assignments and quizzes for a section of ~30 graduate students.
- Developed reproducible Python scripts and datasets implementing machine learning methods for public policy applications.

Universidad del Pacífico – Faculty of Economics and Finance — Lima, Peru
Lecturer (2020–2024) | Teaching Assistant (2015–2019) — **April 2015 – July 2024**

- Taught applied economics and econometrics for 8+ years, guiding students through regression analysis with real-world data.

OTHER PROFESSIONAL EXPERIENCE

Ministry of Foreign Trade and Tourism (Peru) – COPESCO — Lima, Peru
Public Investment Formulation Assistant — **November 2023 – April 2024**

- Monitored evaluation and compliance for 10+ public tourism infrastructure projects, assessing proposals under Peru's Invierte.pe system.
- Cleaned and structured administrative datasets in Excel for feasibility analysis and descriptive statistical reporting.

Ministry of Culture (Peru) – Economics Studies Unit — Lima, Peru
Data Consultant — **October 2020 – February 2021**

- Consolidated 5 quantitative data sources and qualitative inputs into a centralized Excel dashboard for budget and policy analysis.
- Produced analytical tables and visualizations to support budget allocation decisions and policy briefs.

TECHNICAL SKILLS

Programming: Python, SQL, R, Stata, JavaScript (D3.js)

Python Libraries: pandas, scikit-learn, GeoPandas, Shapely, Selenium, Altair, matplotlib

Tools: Git, Jupyter Notebooks, pytest, Excel (cost modeling, Pivot Tables, VBA, dashboards) | Languages: English (Fluent), Spanish (Native)

Methods: Econometrics & causal inference, composite index construction, cost modeling, machine learning, spatial analysis, web scraping & APIs